

Decision Log – Skylark Drones GCP Marker Detection Assignment

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Problem Statement

The objective of this assignment is to develop a machine learning pipeline capable of:

1. Predicting the pixel coordinates (x, y) corresponding to the center of a Ground Control Point (GCP) marker.
 2. Classifying the detected marker into one of the following categories:
 - Cross
 - Square
 - L-Shape
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Dataset Understanding

The provided dataset contains high-resolution aerial drone imagery and annotation data in JSON format.

Each annotation contains:

- Marker center coordinates (x, y)
- Marker shape label

The dataset consists of approximately 1000 annotated images.

Exploratory Data Analysis (EDA)

The following observations were made during analysis:

- Total annotated images: 1000
- Valid labeled samples: 996
- Missing shape annotations: 4
- Image resolution: 4096 × 3068 pixels

- Marker classes:
 - Cross
 - Square
 - L-Shape

Several samples were visualized to understand marker appearance and scale. It was observed that GCP markers occupy a very small portion of the overall image area.

Data Preprocessing

The original annotations contained only marker center coordinates and shape labels.

Since YOLO-based object detectors require bounding box annotations, synthetic bounding boxes were generated around each marker center point.

Preprocessing steps:

1. Load JSON annotations.
 2. Remove invalid samples with missing labels.
 3. Generate fixed-size bounding boxes around marker centers.
 4. Convert annotations into YOLO format.
 5. Create train and validation datasets.
 6. Generate dataset configuration file (dataset.yaml).
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Model Selection

YOLOv8n was selected because:

- Lightweight and efficient architecture.
- Suitable for small-object detection.
- Supports simultaneous localization and classification.
- Fast inference speed.

Model used:

- YOLOv8n (Ultralytics)
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Training Strategy

Training Configuration:

- Framework: Ultralytics YOLOv8

- Model: YOLOv8n
- Image Size: 640 × 640
- Batch Size: 16
- Dataset Split:
 - Training: 80%
 - Validation: 20%

The model was trained to detect GCP markers and classify them into the three target categories.

Challenges Encountered

Small Object Detection

The GCP markers occupy only a small fraction of the image area, making detection challenging.

Missing Labels

A small number of samples contained missing shape annotations and were excluded from training.

Runtime Constraints

GPU quota limitations and runtime interruptions were encountered during experimentation and model training.

Inference Pipeline

The trained YOLOv8 model was used to perform inference on the provided test dataset.

For each detected marker:

1. Bounding box coordinates were obtained.
 2. Marker center coordinates were calculated from the predicted bounding box.
 3. Shape class was extracted from model predictions.
 4. Predictions were exported to JSON format.
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Deliverables

Generated outputs:

- Trained model weights (`best.pt`)
 - Prediction file (`predictions.json`)
 - Source code and notebook
 - GitHub repository
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Future Improvements

Potential improvements include:

- Keypoint-based center localization.
 - Adaptive bounding box generation.
 - Larger YOLO architectures.
 - Additional data augmentation for improved small-object detection performance.
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Conclusion

A complete end-to-end pipeline was developed for GCP marker localization and classification. The solution performs data preprocessing, model training, inference, and prediction export using YOLOv8 and generates the required submission outputs.